

Experiment- 03

Study of Different Types of LAN & Network Equipment

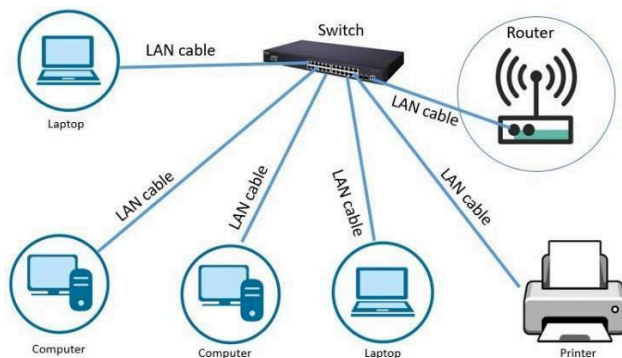
Lab Objective: Understanding various types of LAN (Local Area Networks) and network equipment commonly used in networking environments.

Prerequisites: Basic knowledge of networking concepts and familiarity with common networking devices

Outcome: Gain practical experience in identifying and understanding different types of LAN and network equipment.

Description:

A Local Area Network (LAN) is a network that connects computers and devices in a limited geographical area, such as a home, school, computer laboratory, office building, or closely positioned group of buildings. LANs are characterized by higher data-transfer rates, smaller geographic range, and lack of a need for leased telecommunication lines.



Local Area Network

Types of LANs:

1. **Wired LAN:**
 - o Uses cables and physical hardware (e.g., Ethernet cables).
 - o Commonly uses Ethernet standards (IEEE 802.3).
2. **Wireless LAN (WLAN):**
 - o Uses wireless signals (e.g., Wi-Fi).
 - o Commonly uses Wi-Fi standards (IEEE 802.11).
3. **Virtual LAN (VLAN):**

- o Uses software to group network devices into virtual segments, regardless of physical location.
- o Enhances network security and performance.

Network Equipment:

1. **Router:** Connects different networks and directs network traffic.



2. **Switch:** Connects devices within the same network and uses MAC addresses to forward data.



3. **Access Point (AP):** Allows wireless devices to connect to a wired network.



4. **Network Interface Card (NIC):** Hardware component that connects a computer to a network.

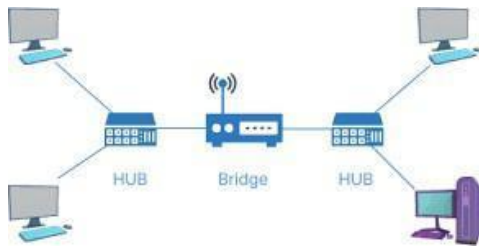


5. **Modem:** Converts digital data to analog signals and vice versa for data transmission over telephone lines.

6. **Repeater:** Amplifies or regenerates signals to extend the distance over which data can travel.



7. **Bridge:** Connects and filters traffic between two network segments.



8. **Firewall:** Monitors and controls incoming and outgoing network traffic based on security rules.



Steps of Installation / Configuration

1. Setting Up a Wired LAN:

- **Step 1:** Gather all necessary hardware (Ethernet cables, switches, routers, NICs).
- **Step 2:** Connect the router to the modem.
- **Step 3:** Connect the switch to the router.
- **Step 4:** Connect computers and other devices to the switch using Ethernet cables.
- **Step 5:** Configure the router settings via its web interface (usually accessed through a web browser).

- **Step 6:** Ensure all devices are configured with appropriate IP addresses, either dynamically (via DHCP) or statically.

2. Setting Up a Wireless LAN:

- **Step 1:** Gather all necessary hardware (Wi-Fi router, access points if needed).
- **Step 2:** Connect the Wi-Fi router to the modem.
- **Step 3:** Access the router's web interface to configure wireless settings (SSID, security settings).
- **Step 4:** Place access points in optimal locations to ensure coverage.
- **Step 5:** Connect devices to the Wi-Fi network using the configured SSID and password.

3. Configuring VLANs:

- **Step 1:** Access the switch's management interface (usually through a web browser).
- **Step 2:** Navigate to the VLAN configuration section.
- **Step 3:** Create new VLANs and assign ports to the respective VLANs.
- **Step 4:** Configure routers or layer 3 switches to route traffic between VLANs if needed.